

PINN–PINO TwinMoE: Physics-Constrained Expert Coordination for Multiscale Battery Digital Twins

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Code and training: [DeepMonkey-Studio / research/battery-twin](#)

Study status. Implementation and recorded experiments; joint expert-policy evaluation and causal forecasting comparisons remain incomplete.

Abstract

Battery digital twins operate across different time scales: degradation develops over cycles, while concentration, voltage, and temperature respond within an operating episode. We study a framework that coordinates a physics-informed degradation expert, a temporal spectral operator, and explicit expert-adoption policies. The long-horizon model couples a predictive backbone to a single-particle physics head. The short-horizon model mixes lightweight and operator experts, with optional conditional execution. Service policies select accepted results using domain validity, confidence, and disagreement while retaining model provenance. In an archived 77-cell HUST holdout, the PINN reduces mean normalized RMSE from 0.166026 to 0.162743 and improves three trajectory-shape metrics, although mean MAPE increases from 14.165708% to 14.753008%. A separate 63-scenario synthetic operator test records voltage MAE of 0.018221 V. These complementary results motivate evaluating accuracy, physical consistency, and execution cost jointly. We define the remaining controlled experiment for common-reference routing and causal forecasting.

1 Introduction

An operational battery twin needs both a view of aging and a response to changing current and temperature. Long-horizon degradation models estimate capacity trajectories and life-related quantities; short-horizon models estimate electrothermal state. The preferred expert may change with available measurements, extrapolation, and the cost of an incorrect result.

Physics-informed neural operators already combine data and differential-equation constraints [1], and physics-informed mixtures have been studied for battery degradation [2]. Our focus is the coordination of different temporal tasks and different levels of expert selection. We distinguish learned mixing inside a model from service rules that decide which result to adopt.

The study covers three implemented components: a degradation backbone with a single-particle physics head; a temporal spectral operator with lightweight and operator experts; and runtime policies that retain alternative predictions and select a primary result. The experiments examine where physical constraints improve trajectories and where prediction error or computational cost remains unfavorable.

Research questions. RQ1: How do physical constraints affect degradation error, trajectory shape, and effective-parameter recovery? RQ2: How does conditional operator execution change error and measured cost under a common counterfactual reference? RQ3: Can expert adoption preserve valid state across domain changes and interrupted requests?

2 Implemented Method

2.1 Long-Horizon Physics-Informed Expert

The implementation augments the BatteryMFormer backbone with **BatterySPMPINNHead**. Cell and measured-curve context inform concentration and effective physical parameters. Its single-particle constraints address radial diffusion, surface flux, conservation, initial conditions, and voltage consistency. A schematic radial diffusion residual is

$$\mathcal{R}_c = \partial_t c - \frac{D_{\text{eff}}}{r^2} \partial_r (r^2 \partial_r c),$$

with coordinate scaling and effective parameters defined by the implementation. Center/surface boundary conditions and integral constraints complement interior collocation. Additional terms include lithium inventory, rest-weighted open-circuit-voltage anchoring, chemistry priors, aging dynamics, and observed physical quantities.

The training objective combines prediction and physical terms. Each physical loss is scaled by its detached current absolute magnitude with a lower clamp. Diagnostics measure the cosine between data and physics gradients on shared parameters; conflicting physics pressure is attenuated toward a configurable minimum. This is loss-weight attenuation, not gradient projection. Effective normalized diffusion parameters require independent geometry and electrochemical information before interpretation as intrinsic material diffusivities.

2.2 Short-Horizon Physics-Informed Operator

SPMPINOTransformer combines a temporal representation with radial spectral blocks and electrothermal outputs. Let $f_L(x)$ and $f_O(x)$ denote lightweight and operator concentration predictions and $w(x) \in [0, 1]$ the learned operator weight. The soft mixture is

$$\hat{c}(x) = (1 - w(x))f_L(x) + w(x)f_O(x).$$

In **dynamic** mode both experts execute. In **dynamic_sparse**, the lightweight expert executes first and the operator runs when physical risk or learned probability exceeds its configured threshold. The native ONNX export currently uses **dynamic**.

The model accepts optional health/degradation context. Inspected training paths construct this context from capacity ratio and current/voltage statistics. The architecture therefore offers a conditioning interface across time scales; a measured handoff from the long-horizon PINN output to the short-horizon PINO remains an experiment to complete. The framework does not assume joint end-to-end training of all experts.

2.3 Three Levels of Expert Coordination

Level	Decision and retained information
Intra-model	Learned light/operator weights; sparse mode also uses physical-risk thresholds.
Life prediction	Rules inspect extrapolation, confidence, domain warnings, and standard/PINN disagreement before selecting the accepted life result.
Twin runtime	Rules select qualified PINO, electrothermal baseline, or physical SPM fallback, retaining alternative predictions and model identity.

The life policy flags high extrapolation, weak fit, or inadequate confidence. Large disagreement requires both an absolute and a relative threshold. The twin runtime extracts bounded terminal SOC and temperature for committed state. Logs distinguish executed experts from the adopted expert, which supports analysis of rejected predictions and fallback costs.

2.4 Deployment

The platform has separate native ONNX model packages for life and transient inference. A Rust host handles the deployed inference path without a Python/PyTorch runtime dependency. Runtime provenance and domain checks accompany predictions. Numerical agreement between exported and research models, request latency, and handoff behavior are separate evaluation targets from predictive accuracy.

3 Data and Evaluation Protocol

The inspected frozen synthetic comparison records 1,255 training, 127 validation, and 63 test scenarios from an LFP electrothermal dataset. These are simulation scenarios, not 1,445 independently measured physical cells. Group assignments, generation parameters, failures, and source-cell identifiers require verification from the dataset manifest before final publication.

The inference task must also be explicit. The comparison script calls an evaluation function whose model wrapper defaults to **observer**. The wrapper’s separate **forecast** mode restricts voltage context to a historical prefix. Thus, the existing observer-path results cannot, by themselves, establish causal future prediction. Whether the selected checkpoint actually consumes the available voltage context must be inspected from its configuration. Final experiments will separately evaluate observer inference and historical-prefix forecasting with a declared prediction cutoff and known-control assumptions.

3.1 A Common Counterfactual Routing Oracle

The existing evaluator computes expert costs from the outputs returned by the current routing mode. In sparse mode, unexecuted operator outputs are set equal to lightweight outputs; the operator cost also receives a fixed positive penalty. This construction makes unexecuted samples favor the lightweight expert in the reference labels.

The revised experiment will evaluate both frozen experts offline for every test scenario and save a common label

$$y^*(x) = \operatorname{argmin}_{e \in \{L, O\}} [\ell_e(x) + \lambda C_e],$$

where ℓ_e combines declared predictive and physical criteria and C_e is a predeclared compute cost. All routing policies will use these labels for evaluation. The reference-label computation will not be counted as online policy cost. We will report predictive error directly in addition to oracle agreement; an oracle score alone is not a physical accuracy measure.

3.2 Required Comparisons

The comparison must freeze expert checkpoints and include always-light, always-operator, fixed mixture, learned soft routing, rule-only routing, and sparse combined routing. Thresholds and cost weights must be selected on validation data. Physics-component and routing-component ablations should use identical data splits and optimization budgets. Report scenario-group uncertainty, optimization-seed variation, batch size, warm-up, device synchronization, CPU/GPU, peak memory, and request P50/P95 latency. Operator-call fraction is a separate metric from speedup.

4 Recorded Evidence

These values are transcribed from existing artifacts; training was not rerun for this draft. The July 2026 frozen comparison uses checkpoint `spm-pino-transformer-lfp-v10-6-speed-first`. It must not be silently combined with later guarded or transfer checkpoints.

Recorded metric	Dynamic	Dynamic sparse
Concentration MAE (source normalization)	0.008161	0.008711
Voltage MAE (V)	0.018221	0.018236
Temperature MAE (°C)	0.159237	0.159237
Mean request latency (ms)	5.955382	5.675562
Request P95 latency (ms)	8.588700	7.843700

Routing-accuracy values are omitted because the archived labels change with execution mode. Request timing contains 189 repeated calls rather than 189 independent scenarios. Recorded batch P95 latency is 7.007 ms for dynamic and 32.240 ms for sparse, showing that request and batch measurements cannot support a blanket speedup claim. The timing environment and protocol need revalidation.

The three-optimization-seed summary records voltage MAE mean 0.017967 V and sample standard deviation 0.001933 V; concentration MAE has coefficient of variation 17.82%. The seed-2026 checkpoint is named speed-first, whereas the additional checkpoints are named full-seed variants. Configuration identity must be checked before treating this table as a strictly controlled seed-only experiment. These rows also do not establish three-seed common-oracle sparse-routing accuracy.

4.1 Degradation and Physical Consistency

The recorded HUST analysis uses 77 cell IDs, 20,000 bootstrap resamples, and six Holm-corrected paired tests. Table 1 reports cell-aggregated results in the source normalization. Its source-level holdout designation still requires a final audit of upstream training and calibration membership. The mean RMSE difference is -0.003283 , with bootstrap 95% interval $[-0.005754, -0.000627]$ and Holm-adjusted paired-test $p = 0.00644$. Each of the 77 cells favors the PINN on the three recorded shape metrics. Mean MAPE instead increases by 0.587300 percentage points. The results show a tradeoff between pointwise relative error and trajectory shape.

Metric	Standard	PINN
RMSE (normalized)	0.166026	0.162743
MAPE (%)	14.165708	14.753008
Monotonicity violation	0.364614	0.337601
Excess RMS	0.006793	0.005608
Smoothness RMS	0.018637	0.015769

Table 1: Recorded HUST cell-level analysis; lower is preferable for these metrics.

Three internal PINN optimization runs record MAPE values of 1.350986%, 1.3208274%, and 1.3223664%; their mean is 1.3313933%, versus the recorded standard model’s 1.2129676%. Removing chemistry conditioning worsens MAPE in all three paired runs, by an average 0.0584 percentage points. The parameter-freezing ablation reports large surface-flux errors, illustrating why prediction error alone is insufficient for evaluating physical identification.

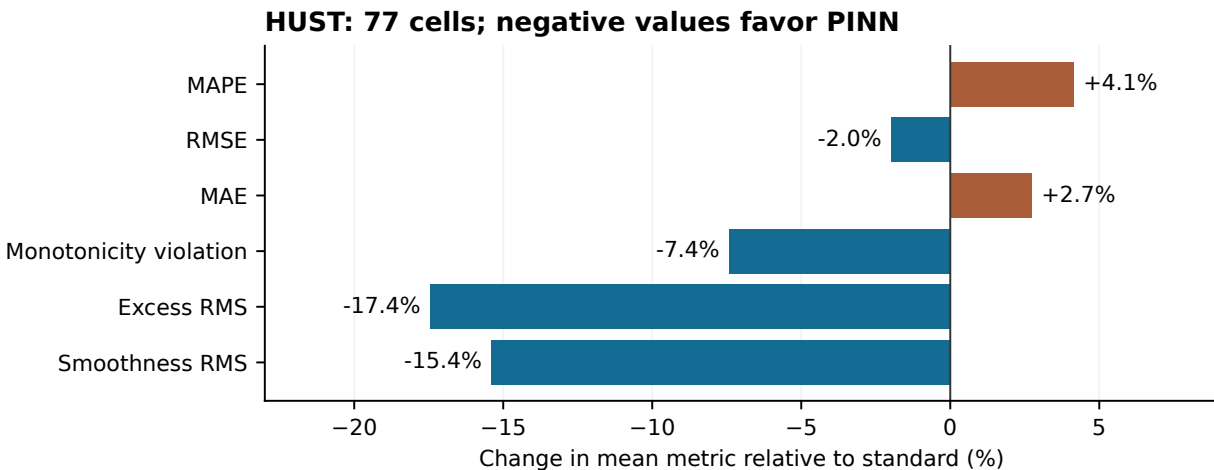


Figure 1: Relative changes computed from the recorded means in the 77-cell HUST analysis. Bars are ratios of aggregate means, not mean per-cell percentage changes; no confidence intervals are implied.

5 Limitations and Remaining Experiments

Synthetic concentration labels do not validate internal concentration fields in measured cells. Near-zero SOC error may reflect shared conservation-based construction rather than sensor-grounded accuracy. Local Jacobian perturbations do not prove global stability. Two large-capacity target cells do not establish broad cross-domain transfer. A one-pack graph residual result that does not beat a simple baseline does not establish pack generalization.

The remaining joint study will compare always-standard, always-physics, rule-only, learned-only, and combined policies on cell-isolated tasks. It will freeze counterfactual expert labels, declare forecast cutoffs, and ablate health context. Paired error and shape metrics will be reported alongside operator calls, rejection rates, state continuity, and synchronized request/batch latency. Measured-voltage targets and source membership must be frozen before final evaluation.

6 Reproducibility and Disclosure

Training source is available in [DeepMonkey-Studio/research/battery-twin](https://github.com/DeepMonkey-Studio/research/battery-twin). The directory contains the PINN physics head, PINO model, training scripts, generated scenarios, tests, and per-file source hashes. Its README provides installation and small training commands. Platform expert adoption is implemented in [batteryExpertRouting.ts](#). Full BatteryMFormer fine-tuning additionally requires upstream adaptations, data, and weights; the published source snapshot is distinct from the archived experimental checkpoints.

Existing evidence is indexed by source path and SHA-256. The reviewed seven-item experiment manifest matches current artifact bytes. Source revision: 501e430095b5 (full hash in the evidence index). Existing local changes make per-file hashes necessary. A public release needs finalized code/data permissions, frozen dependencies, checkpoint hashes, executable evaluation commands, and regenerated result tables.

Generative AI assisted source inspection, organization, and manuscript drafting.

7 Conclusion

PINN–PINO TwinMoE coordinates degradation prediction, transient electrothermal inference, and expert adoption at distinct levels. Existing results show improved degradation shape and normalized RMSE alongside worse relative error, while the operator study quantifies a separate short-horizon inference task. These tradeoffs motivate a common evaluation of accuracy, physical consistency, state continuity, and execution cost across the framework.

References

- [1] Z. Li et al. Physics-Informed Neural Operator for Learning Partial Differential Equations. arXiv:2111.03794. <https://arxiv.org/abs/2111.03794>.
- [2] X. Huang et al. Physics-informed mixture of experts network for interpretable battery degradation trajectory computation amid second-life complexities. arXiv:2506.17755, 2025. <https://arxiv.org/abs/2506.17755>.